

# Effects of Atwood Number on Rayleigh-Taylor Instabilities

by

Michael Lee<sup>1</sup>, Will Emert<sup>1</sup>, Dylan Hakken<sup>2</sup>

as a final project report for

MECH\_ENG 378 — Applied CFD & Heat Transfer  
Dr. Gregory J Wagner<sup>3</sup>

Mar 19, 2025

---

<sup>1</sup> Dept. of Mechanical Engineering, Northwestern University, Evanston, IL, USA

<sup>2</sup> Dept. of Biomedical Engineering, Northwestern University, Evanston, IL, USA

<sup>3</sup> Associate Professor of Mechanical Engineering, Northwestern University, Evanston, IL, USA

# Table of Contents

---

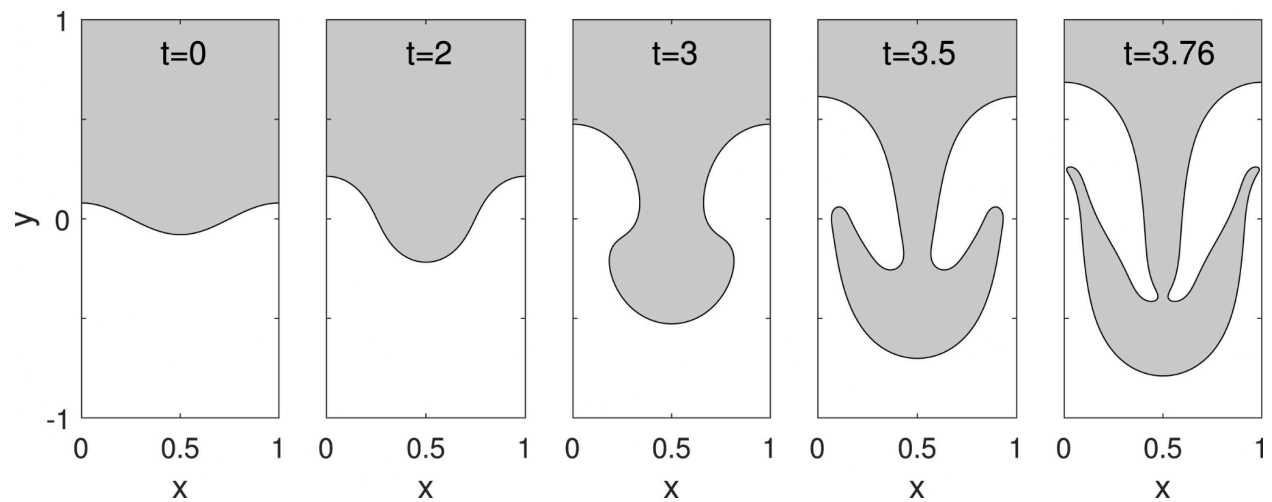
|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>Table of Contents</b>                                        | <b>1</b>  |
| <b>Introduction</b>                                             | <b>2</b>  |
| The Rayleigh-Taylor Instability                                 | 2         |
| Problem Statement                                               | 4         |
| <b>Methods</b>                                                  | <b>5</b>  |
| Geometry and Mesh                                               | 5         |
| Flow Models                                                     | 7         |
| Choosing Solver Scheme & Settings                               | 8         |
| Determining Atwood Numbers & Theoretical Values                 | 9         |
| Data Analysis                                                   | 10        |
| <b>Results</b>                                                  | <b>11</b> |
| Initial Results                                                 | 11        |
| Final Results                                                   | 12        |
| <b>Discussion</b>                                               | <b>16</b> |
| Verification                                                    | 16        |
| Verifying accuracy of FFT                                       | 18        |
| Validation                                                      | 20        |
| Conclusions                                                     | 20        |
| Sources of Error                                                | 20        |
| <b>Appendices</b>                                               | <b>22</b> |
| Appendix A - Visual Comparison of Four Solver Methods           | 22        |
| Appendix B - Graphs from Richardson Extrapolation               | 23        |
| Appendix C - Richardson Extrapolation & GCI Calculations        | 24        |
| Appendix D - 1000/999 Volume Fraction Contours at Various Times | 27        |
| Appendix E - Code for FFT in Python                             | 30        |

## Introduction

---

### The Rayleigh-Taylor Instability

The Rayleigh-Taylor Instability (RTI) is a phenomenon observed when a denser fluid is situated above a less dense fluid. The less dense fluid will push into the denser fluid, creating “fingers”, which are periodic protrusions of the less dense fluid in the denser fluid.<sup>4</sup> That is the laminar regime of this instability. The denser fluid will then accelerate into the less dense fluid and create chaotic turbulent flow that will eventually settle at an equilibrium when the less dense fluid is above the denser fluid.

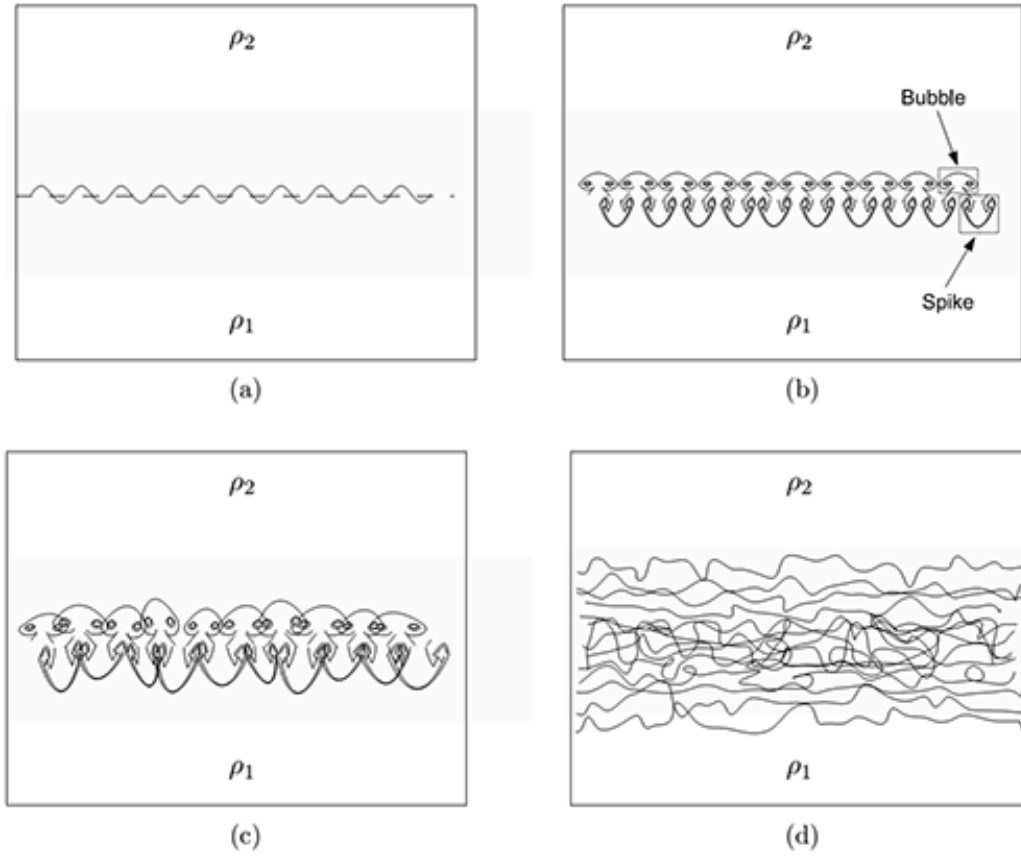


**Figure 1.** This figure shows the development of one finger of the RTI. Our analysis involves development up until the third image, which starts forming mushroom clouds.<sup>5</sup>

The instability is said to have four growth regimes ranging from laminar to turbulent, but our project is concerned with the laminar regime. During laminar flow when the periodic fingers form, the growth rate of the amplitude is linear, and this is called the linear growth regime (Figure 2).

<sup>4</sup> D. H. Sharp, “An Overview of Rayleigh-Taylor Instability,” Los Alamos National Laboratory, May 1983.

<sup>5</sup> S. Shin, S.-I. Sohn, and W. Hwang, “Numerical simulation of single- and multi-mode Rayleigh-Taylor instability with surface tension in two dimensions,” *European Journal of Mechanics - B/Fluids*, vol. 91, pp. 141–151, Jan. 2022.



**Figure 2.** This figure displays the four regimes of the RTI developing from laminar to turbulent flow. We are only analyzing the regime displayed in image (a).<sup>6</sup>

Linear stability analysis can be conducted, which involves multiple equations. First, we can compute the theoretical wavenumber from a given density difference and surface tension as shown in Equation 1.  $\gamma$  is surface tension and  $\rho_2$  is the denser fluid.

$$k_c = \sqrt{\frac{g(\rho_2 - \rho_1)}{\gamma}} \quad (\text{Eq. 1})$$

$k_c$  represents the upper limit of wavenumbers that will incite this instability; wavenumbers above it will be dampened by surface tension. That wavenumber is then used to determine the maximum wavenumber,  $k_m$ , that will dominate when the instability develops a little further (Equation 2.)

$$k_m = \frac{k_c}{\sqrt{3}} \quad (\text{Eq. 2})$$

<sup>6</sup> Roberts, M. S. (2012). “Experiments and simulations on the incompressible, Rayleigh-Taylor instability with small wavelength initial perturbations.”

When it is triggered, multiple wavenumbers will be present at the beginning but the largest one will “win” and become the dominant wavenumber because it grows faster than the other waves. The growth rate of each wave number can be calculated with the theoretical growth rate using Equation 3. The growth rate represents the speed at which the amplitude of the wave is changing.

$$\sigma^2 = \frac{gk(\rho_2 - \rho_1)}{\rho_2 + \rho_1} - \frac{\gamma k^3}{\rho_2 + \rho_1} \quad (\text{Eq. 3})$$

To study RTI under different conditions, we varied the Atwood number, defined in Equation 4. Keeping surface tension and the larger density the same, we assessed how the smaller density affects the development of the RTI by looking at wavenumber and growth rate.

$$At = \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} \quad (\text{Eq. 4})$$

## Problem Statement

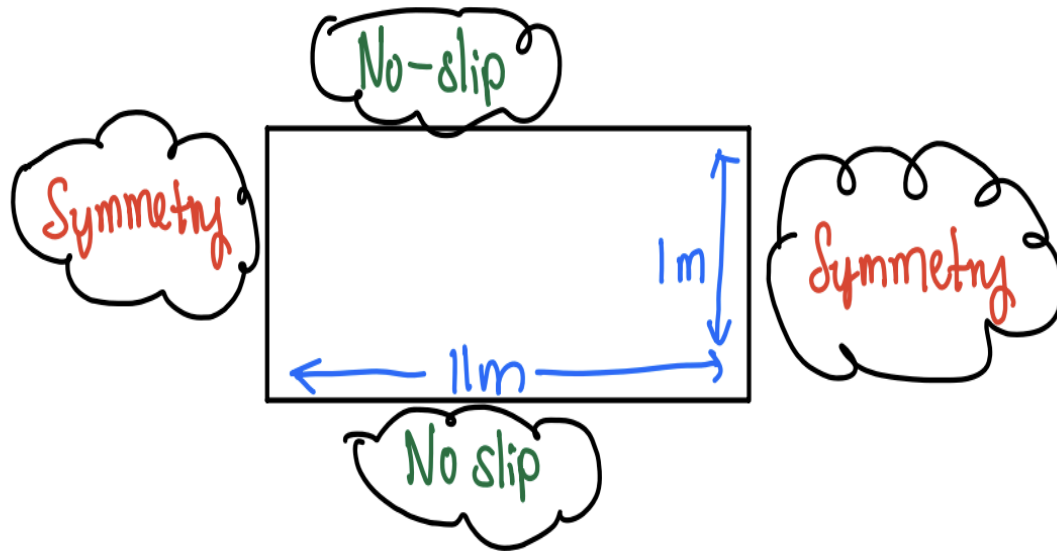
Our problem statement is as follows: It is unclear how the Atwood number affects wavelength and growth rate in the Rayleigh-Taylor Instability model. To address this, we will be comparing theoretical wavenumber and growth rate values to simulated values, and seeing which Atwood number produces the closest to theoretical values. In other words, we are investigating at what Atwood number the simulation starts to become too inaccurate, defined as a greater than 10% deviation from the theoretical values. This deviation was arbitrarily chosen as a cutoff for our analysis. We expect the case with the smallest difference in density to have the closest to theoretical values because the RTI develops clearly when density difference is minimized.

## Methods

---

### Geometry and Mesh

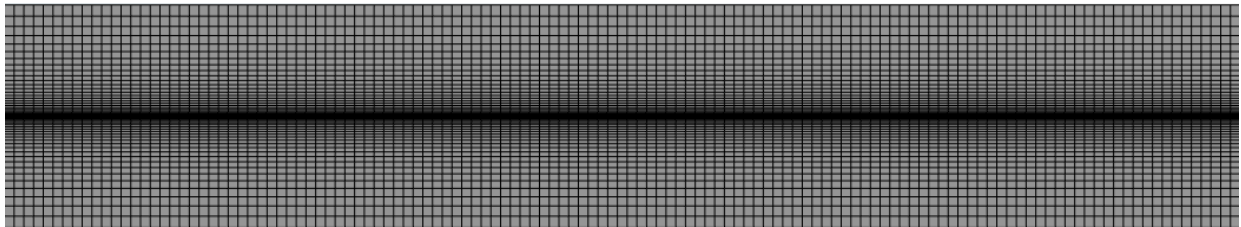
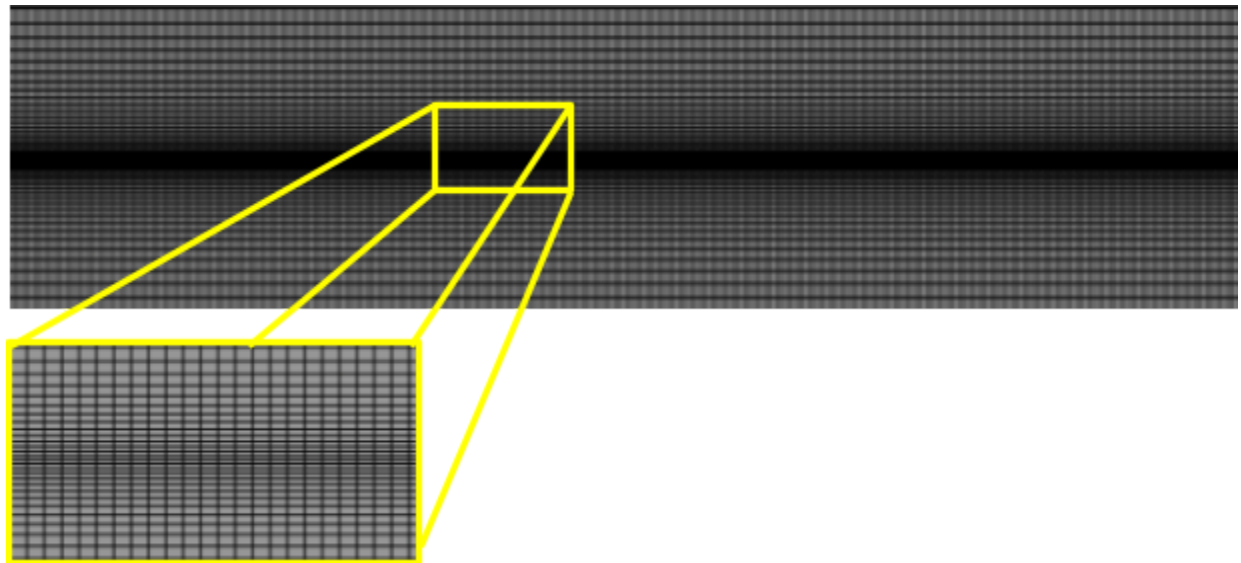
Due to the complexities of this instability, we created a 2D geometry which captured the necessary information about waves at the interface and minimized computational expense. Our geometry was initially an arbitrarily large square with an interface between the two densities exactly in the middle. Because we only need information about the interface, we scaled the height down by 91% to focus on the interface. The height was minimized as much as possible while ensuring we could still capture the amplitude on both sides of the interface. We kept the length the same to maintain a length that was able to include as many waves as necessary. The final dimensions of the geometry were a length of 11 meters and a height of 1 meter. Boundary conditions were symmetry conditions on the vertical sides and no-slip conditions on the horizontal sides. Symmetry conditions were necessary on the vertical sides because this instability technically occurs in an “infinite” domain, meaning there will be no wall effects on the instability. Symmetry provides a slip boundary condition, which does not restrict shear velocity near the wall but still prevents flux through the walls by constraining the normal velocity to be 0. Figure 3 visually demonstrates our geometry and boundary conditions, Table 1 lists the relevant parameters used to create our mesh, and Figures 4-6 visually display the meshes used in Richardson Extrapolation, as explained later. All other simulations used the medium mesh.

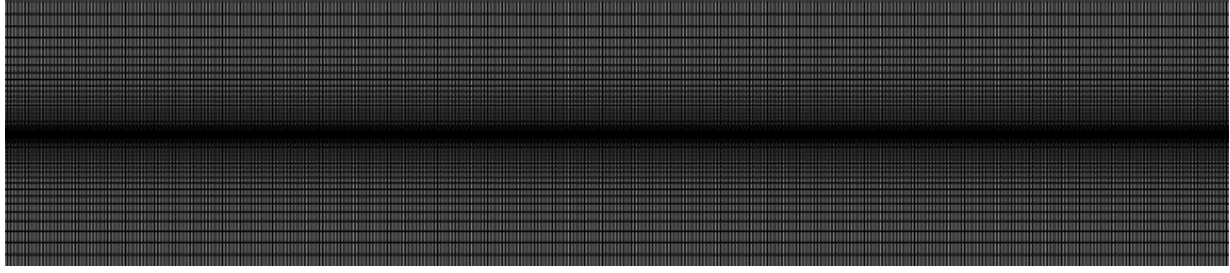


**Figure 3.** Drawing of our 2D geometry. Corresponding boundary conditions applied to each side are labeled.

**Table 1.** Parameters used in mesh.

| Meshing Parameters     |                                                                                      |
|------------------------|--------------------------------------------------------------------------------------|
| Dimensions             | 11m x 1 m rectangle                                                                  |
| Vertical Edge Sizing   | 50 divisions<br>Bias Factor = 10 (— - —)                                             |
| Horizontal Edge Sizing | 250 (coarse), 500 (medium), 1000 (fine) divisions                                    |
| # of (Nodes, Elements) | Coarse, (12801, 12500)<br>Medium (25551, 25000)<br>Fine (51051, 50000)               |
| Others                 | Capture Curvature, Proximity = Off<br>Hard Behavior (sizing) = On<br>Inflation = Off |

**Figure 4.** Coarse mesh**Figure 5.** Medium mesh



**Figure 6.** Fine mesh

## Flow Models

We used a transient simulation with the multiphase VOF (volume of fluid) and inviscid flow to most accurately simulate the RTI. Transient simulations were used because development is dependent on time, and we set gravity as  $-9.81 \text{ m/s}^2$  to replicate Earth's conditions. An arbitrary  $0.1 \text{ N/m}$  (similar to water) surface tension was set as well. The theory of RTI is based on inviscid flow and multiphase was necessary because we were looking at the interface between two different fluids. Additionally, volume of fluid uses the volume fraction equation for conservation of volume fraction and we discuss this method next.

Below is the standard volume fraction equation provided in the ANSYS Fluent theory manual.<sup>7</sup> We can make simplifications by eliminating the right side of the equation: we have no source terms or mass transfer. We can also divide out density because it is constant, resulting in Equation 5. Note that  $\alpha_q$  is the volume fraction ( $0 < \alpha_q < 1$ ) of phase q. In our two-phase simulation, there will be two  $\alpha$ 's.

$$\frac{1}{\rho_q} \left[ \frac{\partial}{\partial t} (\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) \right] = S_{\alpha_q} + \sum_{p=1}^n (\dot{m}_{pq} - \dot{m}_{qp})$$

$$\frac{\partial}{\partial t} (\alpha_q) + \nabla \cdot (\alpha_q \vec{v}_q) = 0 \quad (\text{Eq. 5})$$

Equation 5 tracks the change in alpha over time and the divergence of alpha times velocity. There are two different methods to solve this equation based on how the volume fraction is being discretized: implicit and explicit. We chose to use the implicit method because it can be used with either steady-state or time dependent problems, but the explicit model could also be chosen, as it is said to be more numerically accurate, though time-step size is limited.<sup>7</sup> The implicit discretization equation is shown below and is further simplified to Equation 6 due to the same reasons as Equation 5. This equation is iteratively solved and depends on the current and previous time step to calculate the rate of change of  $\alpha_q$  which should equal the negative product of the volume flux and volume fraction.

<sup>7</sup> ANSYS. (2025). *Chapter 14: Multiphase Flows | 14.3. Volume of Fluid (VOF) Model Theory*. Fluent Theory Guide.



$$\frac{\alpha_q^{n+1}\rho_q^{n+1}-\alpha_q^n\rho_q^n}{\Delta t}V + \sum_f \left( \rho_q^{n+1}U_f^{n+1}\alpha_{q,f}^{n+1} \right) = \left[ S_{\alpha_q} + \sum_{p=1}^n (\dot{m}_{pq} - \dot{m}_{qp}) \right] V$$

$$\frac{\alpha_q^{n+1}-\alpha_q^n}{\Delta t}V + \sum_f (U_f^{n+1}\alpha_{q,f}^{n+1}) = 0 \quad (\text{Eq. 6})$$

Lastly, the inviscid model was used as predictions of the linear growth regime for RTI is based on inviscid flow. Note: although an inviscid model was used for this analysis and the Reynold's number varies due to the acceleration of fluids throughout the simulation, our simulations were in the laminar regime.

### Choosing Solver Scheme & Settings

To start, we ran four simulations with the default settings of each solver scheme: SIMPLE, SIMPLEC, COUPLED, and PISO. We noted that PISO had the most “resolution” and consistency within the periodicity of the interface (see Appendix A). We then performed a more robust, quantitative comparison between PISO and Coupled to resolve a time step resolution issue (discussed later).

For our preliminary comparison, we set up a case with the following parameters:

- **Denser** air on top of **less dense** methane
- 20 iterations per time step (dt), 250 total time steps
- dt = 0.01 s

Upon running our subsequent simulations with PISO, we discovered large errors in wavelength and growth rate (see Results section) and suspected that our large time step sizes were the culprit. However, using any time step below dt = 0.5 s would cause our continuity residual to immediately drop to the 10<sup>-16</sup> range. The simulation would continue through all the time steps and we saw no evidence of a wave developing at the interface between the fluids. We decided to try the Coupled solver scheme and compare against PISO according to Professor Wagner's advice. The setup for this comparison is as follows:

- **Denser** 1000 kg/m<sup>3</sup> material on top of **less dense** 997.5 kg/m<sup>3</sup> material
- 20 iterations per time step (dt), 1500 total time steps
- dt = 0.1 s

We saw residuals that started around the 10<sup>-6</sup> range this time for Coupled and the interface did develop. The Coupled solver produced a simulation with 10% less error in growth rate than PISO for the 1000/997.5 case. Then, we ran our fast fourier transform (FFT) analysis for both a quad and triangular mesh case with roughly the same elements (~25,000 for both) to see which would yield growth rates closer to the RTI linear model's predictions, and quad mesh performed better.

Finally, we fine-tuned the spatial discretization and transient formulation settings based on knowledge that QUICK and Bounded Second Order Implicit for higher accuracy. Using the Bounded Second Order Implicit setting along with QUICK decreased the error of growth rate in the 1000/997.5 case by 11%. The Bounded Second Order Implicit setting without QUICK yielded a similar improvement, but slightly less, so we continued with both settings.

**Figure 7.** Summary of settings used in our final simulations.

## Determining Atwood Numbers & Theoretical Values

Our first “real case” after choosing the models was the  $\rho_2 = 1000 \text{ kg/m}^3$  and  $\rho_1 = 999 \text{ kg/m}^3$  (we will refer to it as the 1000/999 case.) Using Equation 4, this yielded an Atwood number of  $5.00 \times 10^{-4}$ . The corresponding predicted wavelength of this is 1.099 m using Equations 1 and 2 and then converting from the wavenumber to wavelength by dividing  $2\pi$  by the wavenumber. We then decided to use this wavelength as a starting point and varied each case’s wavelength by 0.1 m less than the previous one as shown in Table 2. By fixing  $\rho_2 = 1000 \text{ kg/m}^3$  as the denser fluid, we also calculated the corresponding  $\rho_1$  for each case using Equations 1 and 2, once again. Then, the Atwood number was calculated for each case (Table 2.)

**Table 2.** RTI linear model predictions for each case.

| Case | Predicted wavelength (m) | Calculated $\rho_2$ ( $kg/m^3$ ) | Corresponding Atwood # | Predicted $k_m$ (max wave #) | Predicted Sigma (m/s) |
|------|--------------------------|----------------------------------|------------------------|------------------------------|-----------------------|
| 1    | 1.0988                   | 999.0000                         | 0.00050                | 5.7184                       | 0.1368                |
| 2    | 0.9000                   | 998.5095                         | 0.00075                | 6.9813                       | 0.1845                |
| 3    | 0.8000                   | 998.1136                         | 0.00094                | 7.8540                       | 0.2202                |
| 4    | 0.7000                   | 997.5361                         | 0.00123                | 8.9760                       | 0.2691                |
| 5    | 0.6000                   | 996.6464                         | 0.00168                | 10.4720                      | 0.3392                |

## Data Analysis

To extract wavenumber and growth rate data from the simulations, we ran an FFT (see Appendix E for python script.) Based on 11 time points each 1 second apart, a graph was plotted of time against the natural log of amplitude. This relationship is linear in the linear growth regime, so simply fitting a line is satisfactory. The slope of the line indicates the rate of change of the amplitude, and therefore the growth rate. Wavelength varied between time points so to choose, we used the wavelength at the time point where the dominant wavelength started to take over.

## Results

### Initial Results

Initial results were presented at the final presentation (Table 3.) Case 1 was expected to have the closest to predicted values but had the largest error. We are unsure why this is the case, as we used the same flow and solver schemes for each simulation. Potential sources of error are discussed later. Based on the results, it is difficult to conclude any trends or at what Atwood number the simulation becomes too inaccurate. Assessing results using our 10% error cutoff, only the wavelength for case 2, 3, and 5 are sufficiently accurate. Growth rates tended to have a much larger error than wavelengths. We decided to rerun the simulations using a different solver scheme to hopefully better replicate predicted values.

**Table 3.** Initial results using PISO scheme, Second Order Upwind momentum, First Order Implicit Transient Formulation (as presented in final presentation). All cases ran on medium mesh.

| Case ->                      | 1             | 2             | 3             | 4             | 5             |
|------------------------------|---------------|---------------|---------------|---------------|---------------|
| $\rho_1 (kg/m^3)$            | 999.0000      | 998.5095      | 998.1136      | 997.5361      | 996.6464      |
| Predicted wavelength (m)     | 1.0988        | 0.9000        | 0.8000        | 0.7000        | 0.6000        |
| Predicted growth rate (m/s)  | 0.1368        | 0.1845        | 0.2202        | 0.2691        | 0.3392        |
| Simulated wavelength (m)     | 0.8850        | 0.9080        | 0.8790        | 0.5810        | 0.5490        |
| Simulated growth rate (m/s)  | 0.0780        | 0.1437        | 0.1687        | 0.1726        | 0.2216        |
| <b>% error (wavelength)</b>  | <b>19.46%</b> | <b>0.89%</b>  | <b>9.88%</b>  | <b>17.00%</b> | <b>8.50%</b>  |
| <b>% error (growth rate)</b> | <b>42.97%</b> | <b>22.13%</b> | <b>23.39%</b> | <b>35.86%</b> | <b>34.66%</b> |

### Final Results

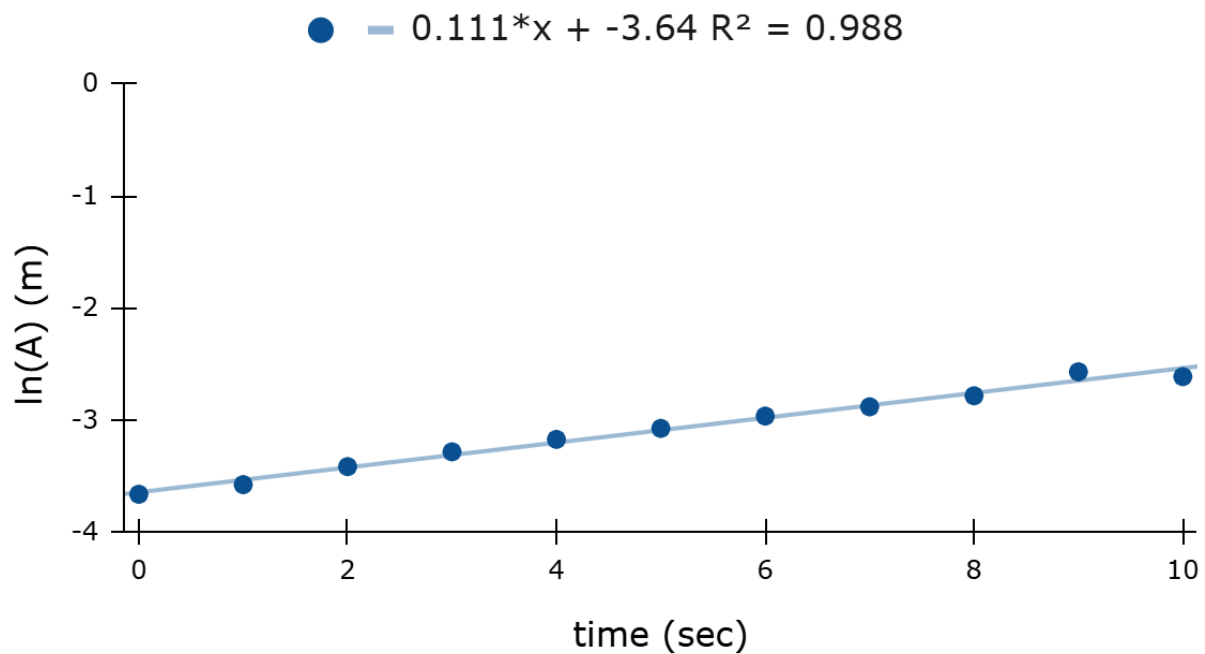
Results were calculated after the final presentation using updated settings that improved performance in the simulations (Table 4.) Case 1 performed better than in the initial results, and each of our predicted growth rates for all the cases were much closer to the simulated results. The deviation in our wavelength appears about the same between the initial results and final results, but if we use our 10% cutoff error on the wavelength, only cases 1 and 2 are sufficiently accurate. This means that even though our error for the growth rate decreased significantly across all cases, the error in our wavelength increased as a result. We found these growth rates from Figures 8-12, where the growth rate is the slope of each of the lines.

**Table 4.** Final results using Coupled VOF scheme with QUICK momentum, Bounded Second Order Implicit Transient Formulation. All cases ran on medium mesh. See figures following Table 4

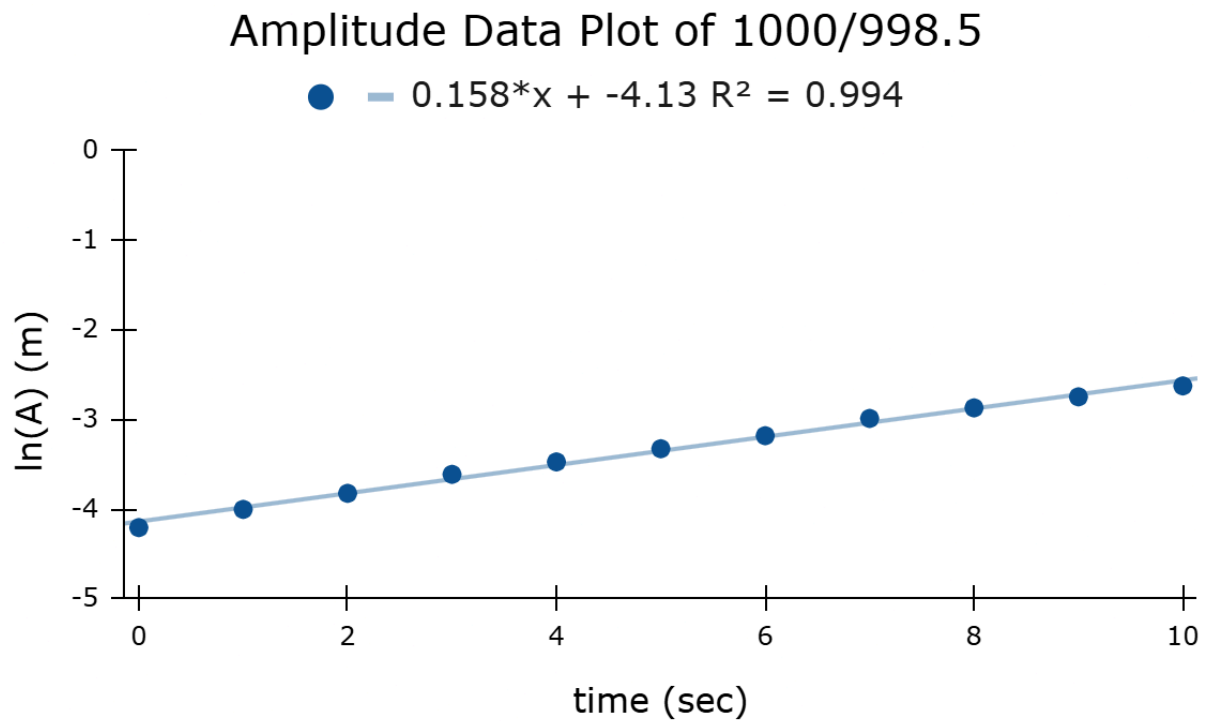
for the corresponding graphs of 11 time points as mentioned in the Methods section for calculating the growth rate (slope).

| Case ->                       | 1        | 2        | 3        | 4        | 5        |
|-------------------------------|----------|----------|----------|----------|----------|
| $\rho_1$ (kg/m <sup>3</sup> ) | 999.0000 | 998.5095 | 998.1136 | 997.5361 | 996.6464 |
| Predicted wavelength (m)      | 1.0988   | 0.9000   | 0.8000   | 0.7000   | 0.6000   |
| Predicted growth rate (m/s)   | 0.1368   | 0.1845   | 0.2202   | 0.2691   | 0.3392   |
| Simulated wavelength (m)      | 0.9940   | 0.9340   | 0.6770   | 0.6300   | 0.5270   |
| Simulated growth rate (m/s)   | 0.1111   | 0.1575   | 0.1867   | 0.231    | 0.2478   |
| % error (wavelength)          | 9.54%    | 3.78%    | 15.38%   | 10.00%   | 12.17%   |
| % error (growth rate)         | 18.77%   | 14.65%   | 15.22%   | 14.15%   | 26.94%   |

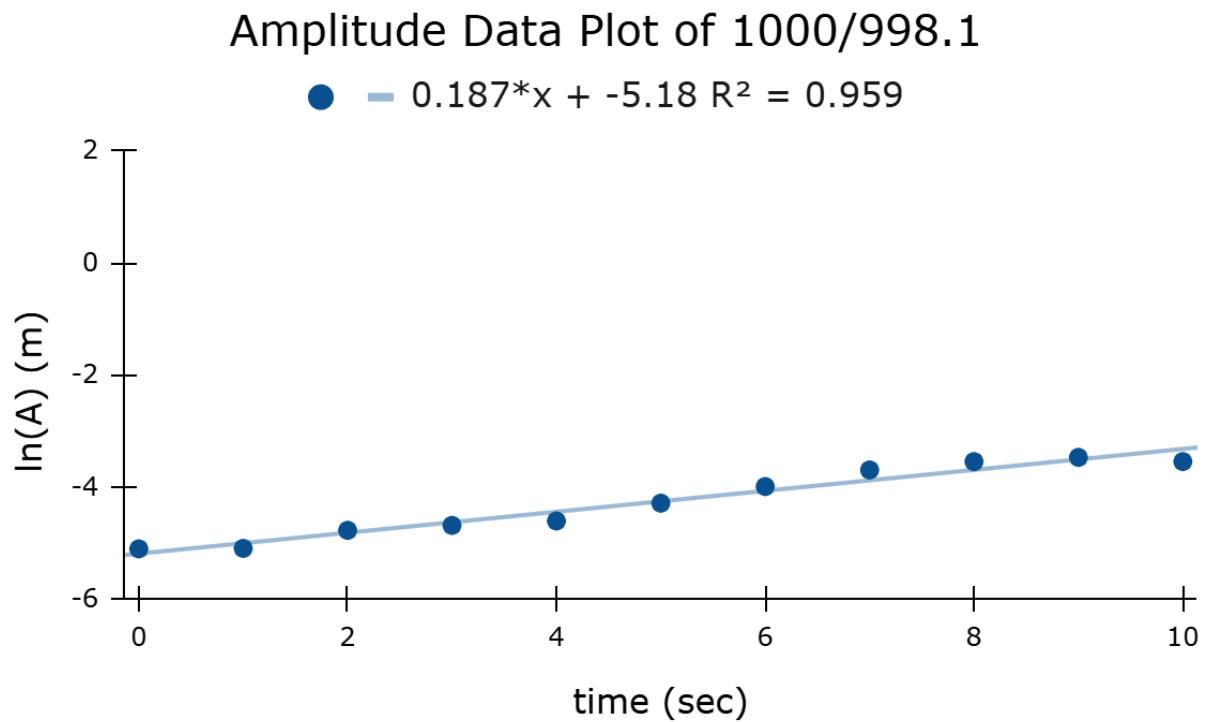
### Amplitude Data Plot of 1000/999 - Medium Case



**Figure 8.** 1000 over 999 case. Sigma is approximately 0.111.



**Figure 9.** 1000 over 998.5 case. Sigma is approximately 0.158.



**Figure 10.** 1000 over 998.1 case. Sigma is approximately 0.187.

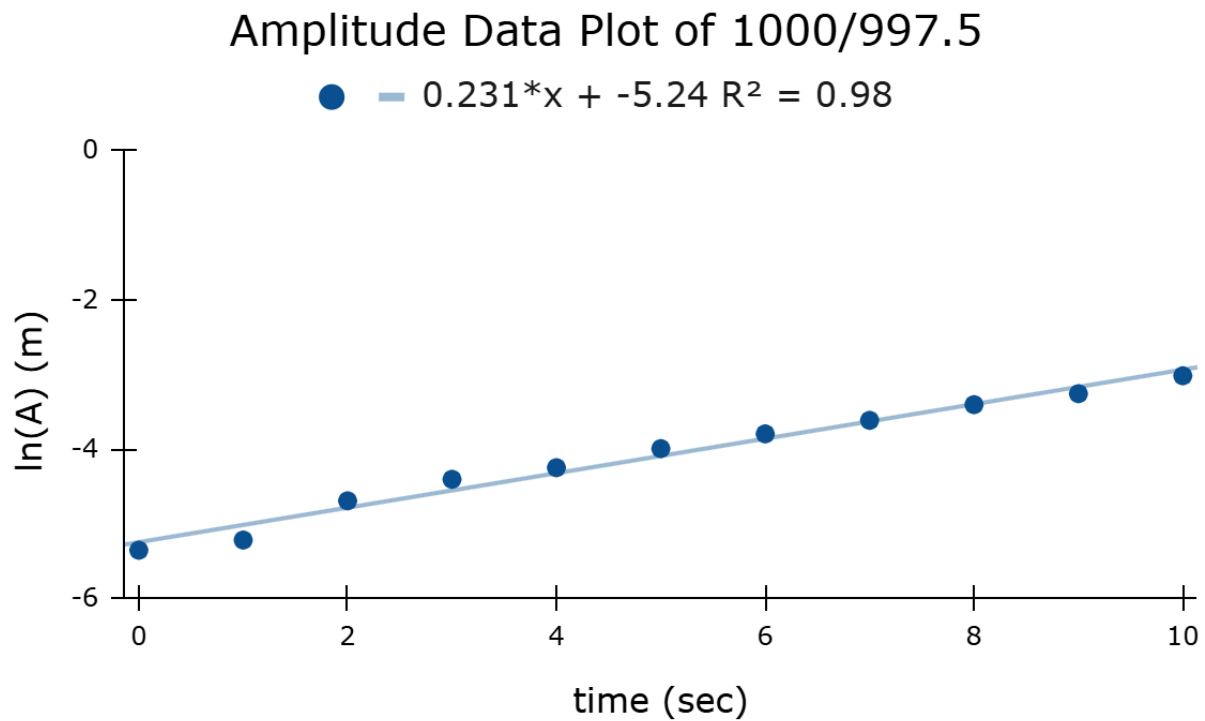


Figure 11. 1000 over 997.5 case. Sigma is approximately 0.231.

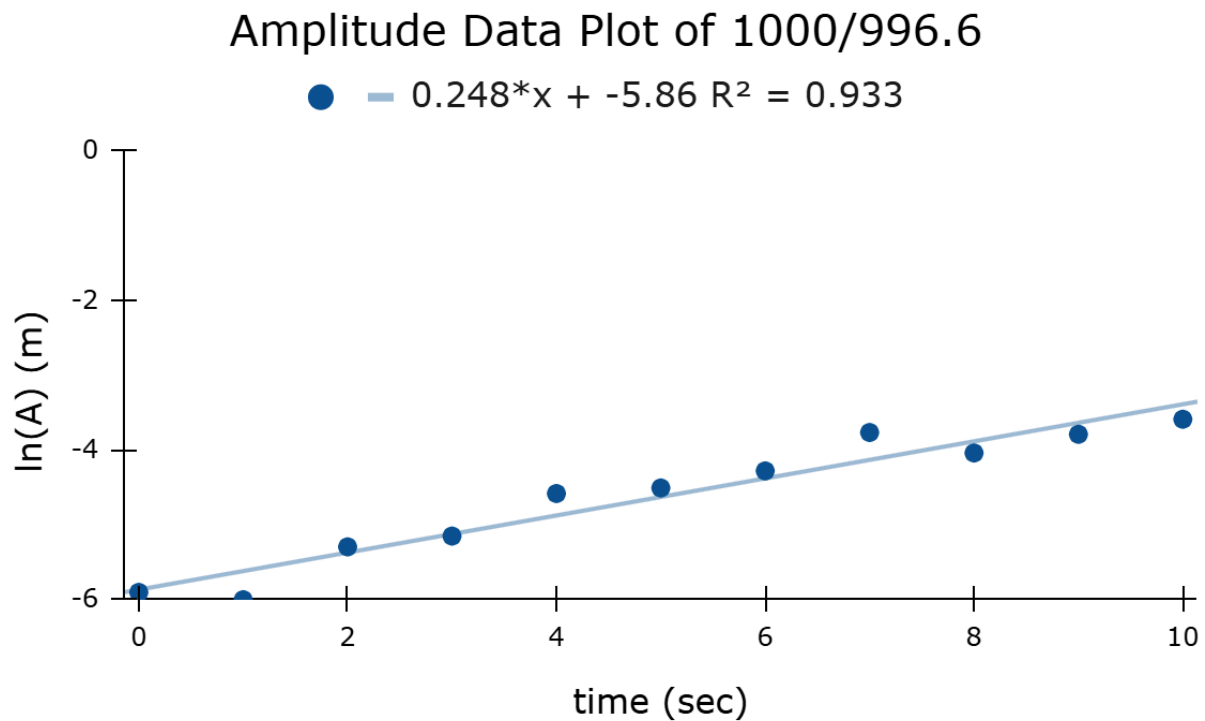


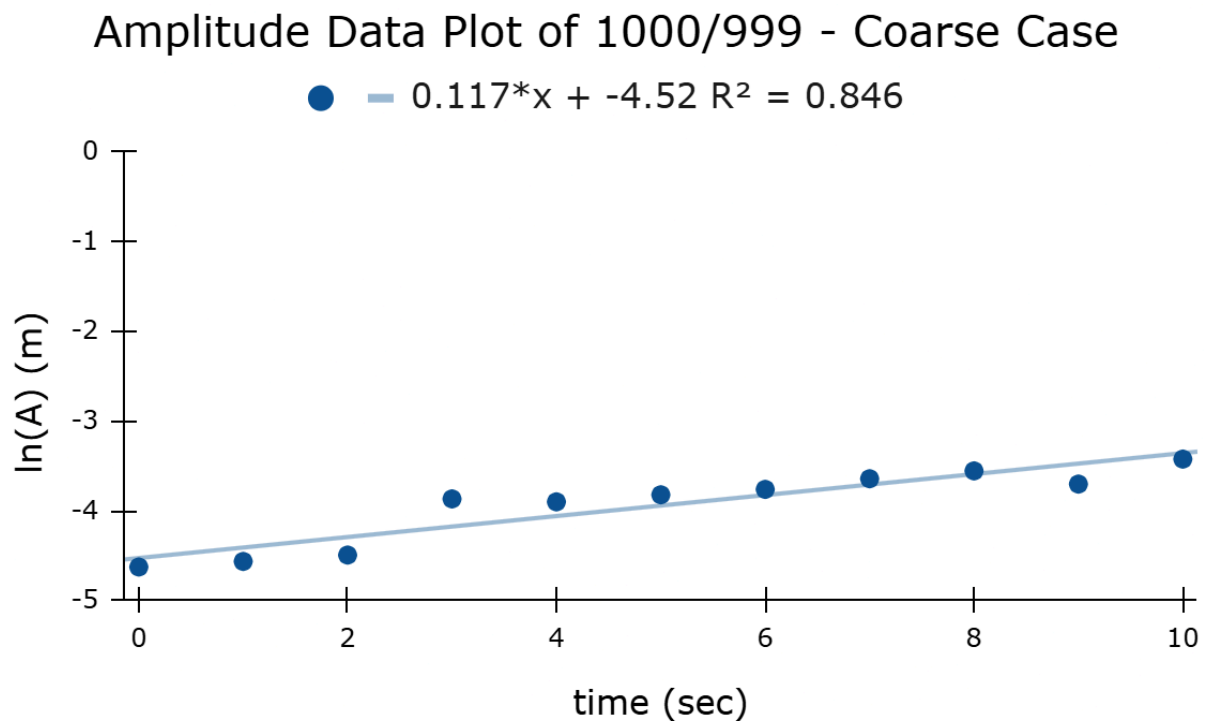
Figure 12. 1000 over 999 case. Sigma is approximately 0.248.

## Discussion

---

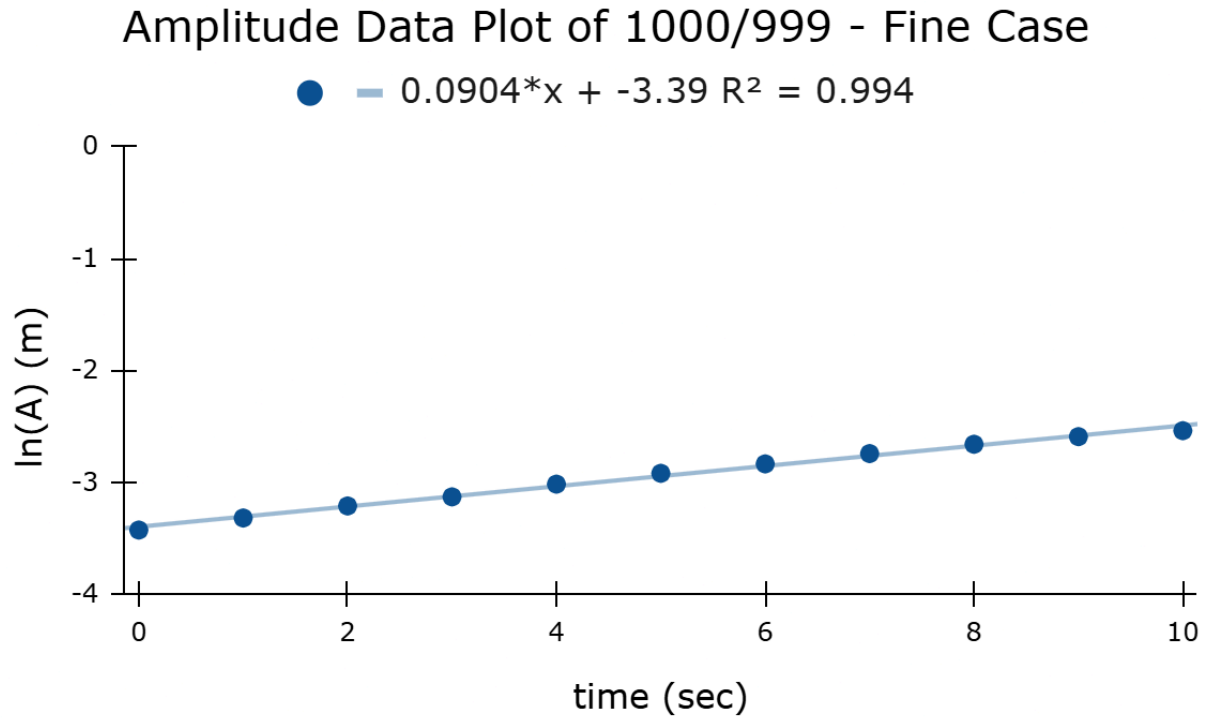
### Verification

We used Richardson Extrapolation on one representative case (1000/999) with the three mesh resolutions to determine our simulation's dependency on mesh size. Additionally, GCI (grid convergence index) was calculated for associated values. Richardson Extrapolation is used to estimate the first-order error  $e = |u_{\text{experimental}} - u_{\text{exact}}|$  and calculate an approximate  $u_{\text{exact}}$ . GCI was used to ascertain the % error. See Appendix B for more information on these simulations, and Appendix C for more details on how we calculated this in a spreadsheet. Figures 13 and 14 were used in addition to the medium case to determine the  $u1$ ,  $u2$ ,  $u3$  (i.e. sigma of coarse, sigma of medium, sigma of fine meshes, respectively).



**Figure 13.** 1000 over 999 COARSE case. Sigma is approximately 0.117.





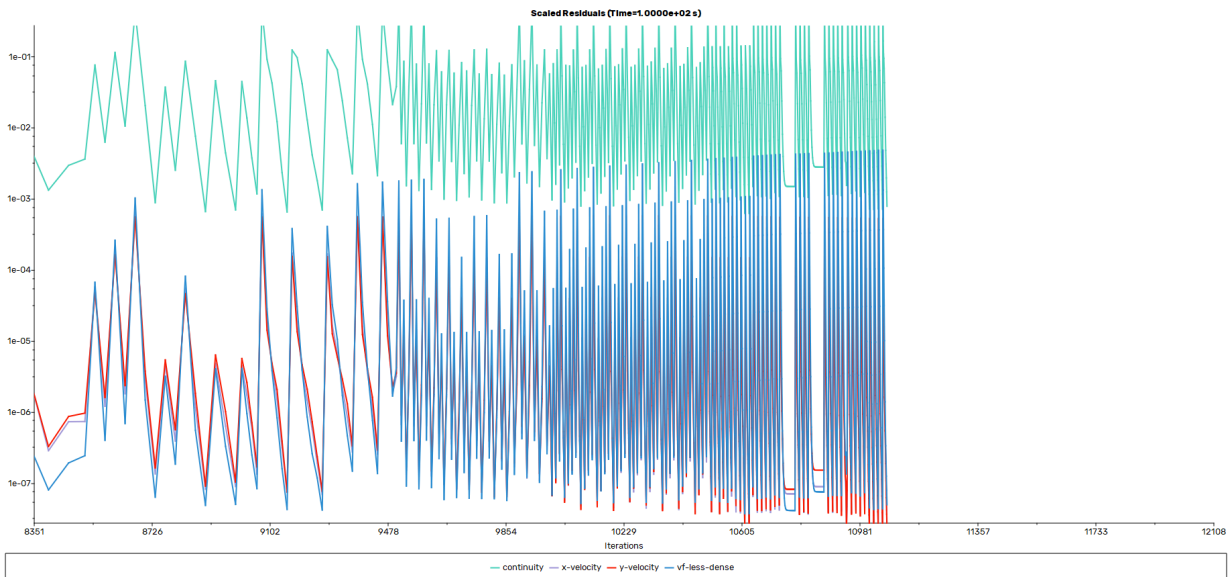
**Figure 14.** 1000 over 999 case. Sigma is approximately 0.0904.

According to Table 5, we see that the Richardson Extrapolated growth rate is approximately 0.1189 m/s with a GCI error of 2.09% among the coarse, medium, and fine meshes. The % error for this growth rate (sigma) is approximately 13.11 %.

The extrapolated value for growth rate from Table 5 of 0.1189 m/s (corresponding GCI 2.09%) and its % error of 13.11% is 5.66% lower than the 18.77% (error of growth rate for the 1000/999 case) from Table 4 with the same case on medium mesh by itself. This significant difference from Richardson Extrapolation suggests that there is a dependence on mesh size and the accuracy of our results. In fact, this is evident in our volume fraction contour plots for this 1000/999 case across the three meshes. As shown in Appendix D, we see that going from a coarser to finer mesh leads to less “smearing” of the contour lines near the interface, leading to a more accurate resolution of the interface. Of course, this accuracy is carried through to our final analysis with our 11 picked time points to calculate growth rate from.

**Table 5.** Richardson extrapolated sigma ( $u_{\text{extrap}}$ ) and corresponding GCI for the 1000/999 case.

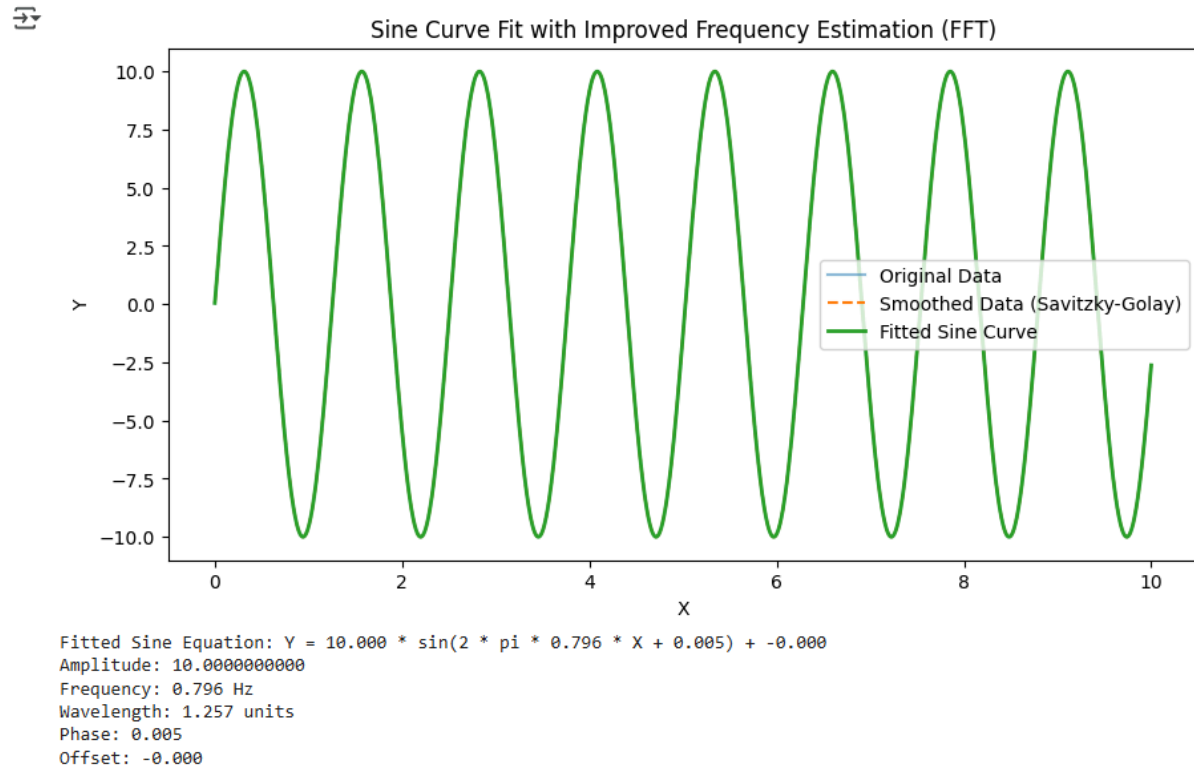
|               |         |                        |         |
|---------------|---------|------------------------|---------|
| <b>u1</b>     | 0.1169  | <b>r_21</b>            | 2       |
| <b>u2</b>     | 0.1106  | <b>r_32</b>            | 2       |
| <b>u3</b>     | 0.0904  | <b>p</b>               | 2.07257 |
| <b>eps_21</b> | -0.0063 | <b>u_extrap</b>        | 0.1189  |
| <b>eps_32</b> | -0.0265 | <b>GCI %</b>           | 2.09%   |
| <b>s</b>      | 1       | <b>% Error (sigma)</b> | 13.11%  |

**Figure 15.** Residuals plot of the 1000/999 case on fine mesh case near  $t=100s$ . Note that the sawtooth pattern is a feature of transient simulations. At each time step, the residuals “jump” up and lower with each subsequent iteration until convergence criteria are met.

## Verifying accuracy of FFT

We decided to plug a sine wave with a known amplitude and wave number into our FFT code to verify the accuracy of the result it was outputting. Below is the script we wrote in matlab to generate the .csv file used, as well as the results of plugging it into our python FFT function. We chose an amplitude of 10 and a wave number of 5.

```
t = linspace(0,10,10000);
x = 10*sin(5*t);
M = [t', x'];
writematrix(M, 'sineWave10Amplitude5WaveNumber.csv')
```



**Figure 16.** Our FFT matches the exact amplitude and nearly exact wave number of the generated data, verifying our FFT code for use with our growth rate prediction graphs.

We got an exact amplitude of 10 and a nearly identical wave number  $2 * \pi * 0.796 = 5.00141$ . This seems to indicate that our FFT code is generating good results because it gave back what was predicted of it. See Appendix D for the final code used in our FFT analysis.

## Validation

As displayed in results, we compared our simulated wavenumbers and growth rates to predicted values. The largest percent error was 27 for growth rate in the 1000/996 case, which we expected to have the largest error because the density difference is the largest. Overall, the simulated wavenumbers and growth rates were not unacceptably large and within an order of magnitude of predicted values, indicating that we chose acceptable simulation settings, with room for improvement.

## Conclusions

Our results indicated two trends when performing tests with fluids close enough in density, and these trends followed what was predicted by equations we derived from Rayleigh-Taylor instability theory. The first correlation was that as the atwood number increased, the simulated wavelength decreased. The second correlation was that as the Atwood number increased, the growth rate increased. We cannot conclude, however, that the error in simulated growth rate and wavelength increased as the difference in density increased.

## Sources of Error

This section discusses potential sources of error and how they could have skewed our results and been avoided.

### Source #1: Implicit vs. explicit volume fraction solver

As mentioned earlier, we chose to use the implicit volume fraction solver over the explicit one but upon further investigation, the explicit volume fraction solver may have been the better choice. Although implicit can be used for transient simulations, explicit has a higher numerical accuracy and we would have preferred to see simulation results more closely mimicking our predicted values. Even if we continued using implicit, it is said to introduce more numerical diffusion with first order transient formulation, so it is necessary to use a higher-order transient formulation. This is another parameter that could be evaluated when choosing methods.

### Source #2: Choice of which time steps to export interface XY position data for

We did not know how to automate the extraction of the position points along the perturbed interface **for all time steps**. Instead, we went through each time step in CFD-Post, chose the time step that we determined to look like the start of the linear regime (smooth sine wave) and took that, along with the 11 subsequent time steps to export XY position data, fit to a sine wave with FFT, and perform our experimental amplitudes, wavelengths/wavenumbers, and growth rates. Ideally, we would extract XY position data from all time steps, run FFTs on all of them, plot results, and use a quantitative way to identify the linear regime to calculate the growth rate. This would give us a larger sample size of time points and hopefully more accurate growth rates, and eliminate the variation in what 11 time points we based the growth rate off of. We did, however, verify the time points picked were in the linear growth regime by plotting time against the natural log of amplitude.

### Source #3: Mesh errors

As explained in the verification section, Richardson Extrapolation confirmed that our simulations were not insignificantly dependent on mesh size, which we realized after completing simulations. We assumed 25,000 elements to be adequate, but this was not the case. It would seem that the number

of elements and element size would more greatly impact later development of the instability, but it impacted early development as well. This is possibly our biggest source of error, and we expect the most improvement of error if we used a smaller element size.

**Below are errors we consider to be less significant, although not negligible.**

Source #4: Time step size

Due to time and computational expense, we were unable to pick time steps smaller than 0.1 seconds. It is possible that a better resolution in time would produce a more accurate development of the RTI, but we thought 0.1 seconds was sufficient. In the future, we could confirm that a maximum of 0.1 seconds is an adequately small time step size by running simulations with smaller time steps.

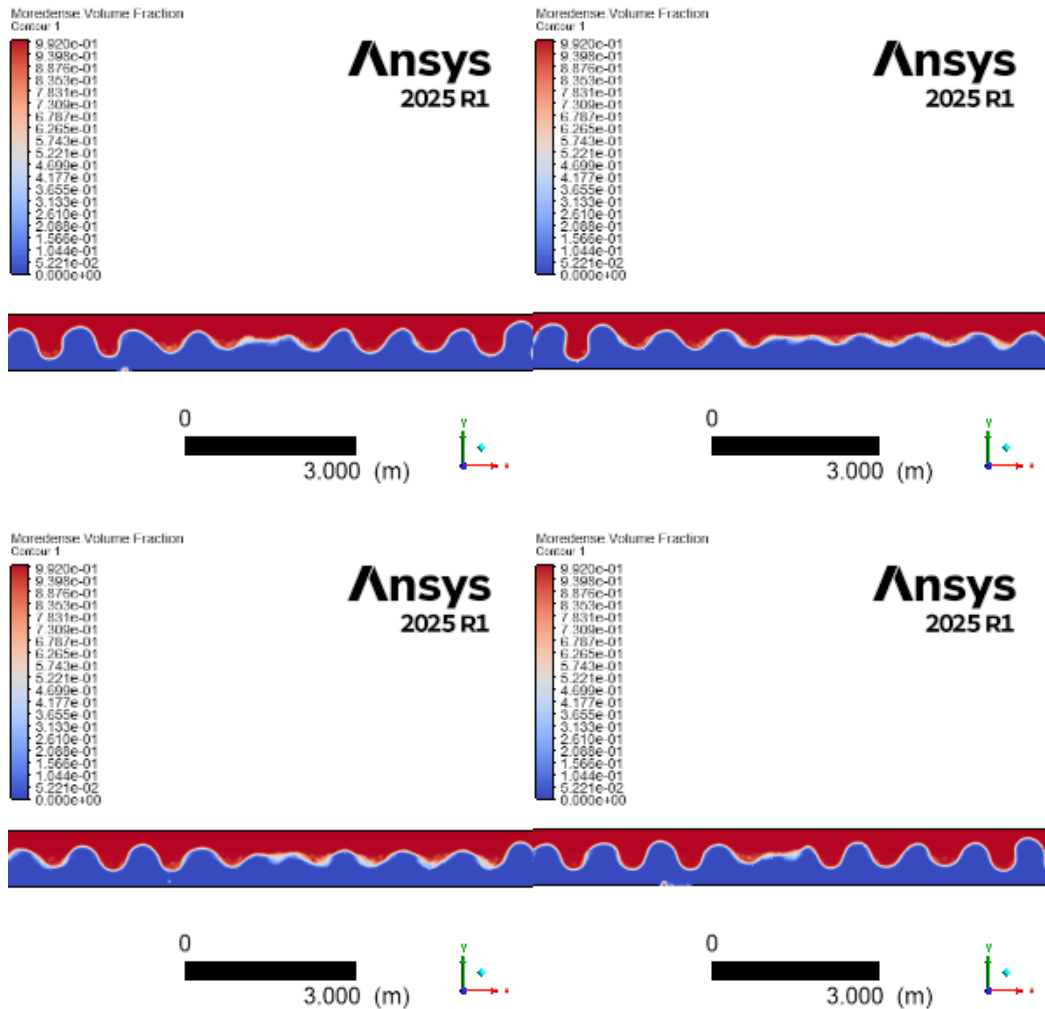
Source #5: Boundary conditions

Because we chose to decrease the dimensions of the box in the vertical direction in order to improve simulation performance, we should have used symmetry boundary conditions on the horizontal walls. Using wall conditions meant that the no slip-condition acted much faster on the walls and could have caused the increase in error in the simulated wavelengths due to dampening of the x-velocity as flow got closer to the wall.

## Appendices

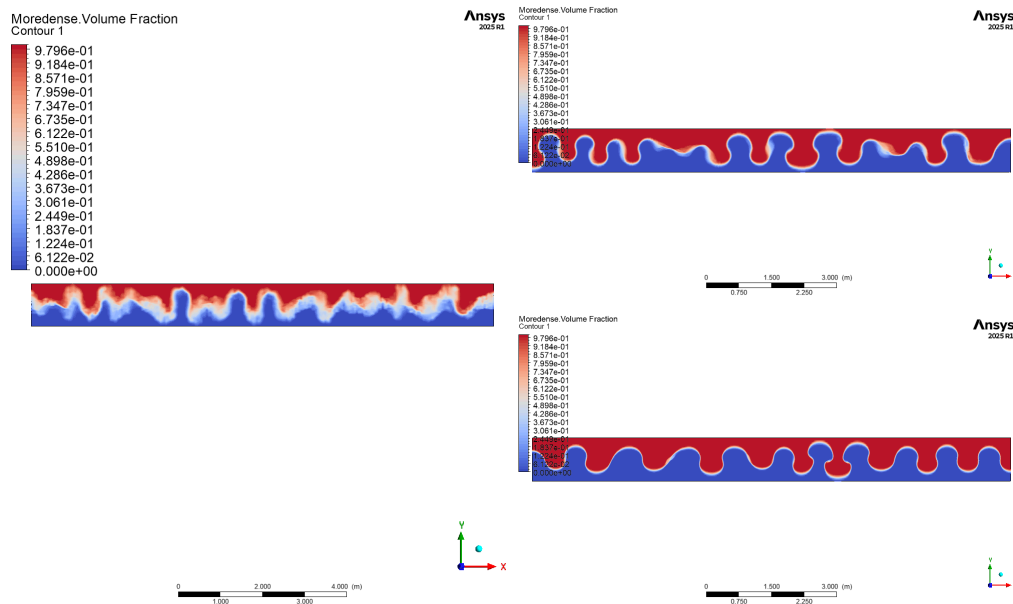
### Appendix A - Visual Comparison of Four Solver Methods

There are less blurry patches along the interface in PISO compared to all other schemes. Note the right most “tear drop” for SIMPLE, second “tear drop” from the left for SIMPLEC, and right most “tear drop” for Coupled result in an amplitude that is larger than the rest of the curve (i.e. it droops below the troughs of the rest of the troughs.)

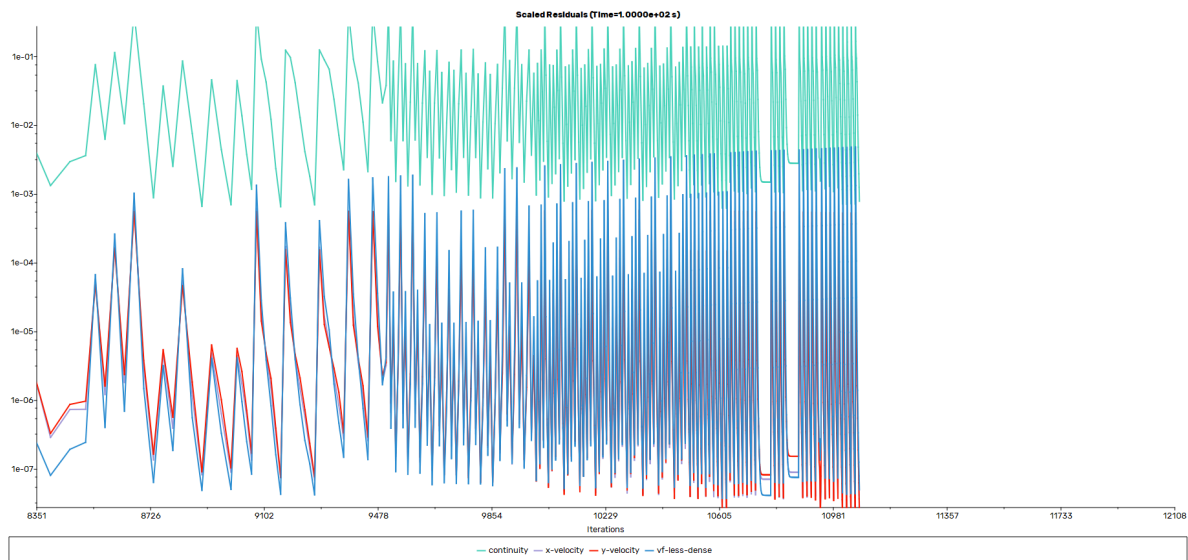


**Figure 17.** From left to right, top to bottom, volume fraction contours of described setup generated with SIMPLE, SIMPLEC, COUPLED, and PISO after 250 time steps ( $dt=0.01s$ )

## Appendix B - Graphs from Richardson Extrapolation



**Figure 18.** 1000/999 volume fraction contours at  $t=100s$  for coarse, medium, and fine meshes (from right to left, top to bottom). See Appendix C for contours at earlier times.



**Figure 19.** Residuals plot of the 1000/999 case on fine mesh case near  $t=100s$ . Note that the sawtooth pattern is a feature of transient simulations. At each time step, the residuals “jump” up and lower with each subsequent iteration until convergence criteria are met.

## Appendix C - Richardson Extrapolation &amp; GCI Calculations

## Richardson Extrapolation

$$e_h = |u_h - u_{exact}| \approx Ch^p.$$

- 3 unknowns:  $p, C, u_{exact}$
- Run with 3 mesh sizes to get 3 equations:

$$\begin{aligned} u_1 - u_{exact} &= Ch_1^p \\ u_2 - u_{exact} &= Ch_2^p \\ u_3 - u_{exact} &= Ch_3^p \end{aligned}$$

where  $h_3 > h_2 > h_1$

- Solve for unknowns to approximate  $u_{exact}$

## Richardson Extrapolation: General Case

- If  $p$  is unknown, solve on 3 grids ( $h_1, h_2, h_3$ ):

$$p = \frac{1}{\ln(r_{21})} \left| \ln \left| \frac{\varepsilon_{32}}{\varepsilon_{21}} \right| + q(p) \right|$$

$$q(p) = \ln \left( \frac{r_{21}^p - s}{r_{32}^p - s} \right)$$

$$s = \operatorname{sgn} \left( \frac{\varepsilon_{32}}{\varepsilon_{21}} \right)$$

where  $r_{21} = \frac{h_2}{h_1}$ ,  $r_{32} = \frac{h_3}{h_2}$ ,  $\varepsilon_{21} = u_2 - u_1$ ,  $\varepsilon_{32} = u_3 - u_2$

- Extrapolated solution:

$$u_{extrap} = u_1 + \frac{u_1 - u_2}{r_{21}^p - 1}$$

- See paper by Celik et al., 2008, J Fluids Eng. **130** for full details

**Figure 20.** Slides from Dr. Wagner<sup>8</sup> on Richardson Extrapolation.  $e_h$  is the first-order approximation of the error.

<sup>8</sup> Wagner, G. J. (n.d.). *Lecture 5: Errors and Verification* [ME 378: APPLIED COMPUTATIONAL FLUID DYNAMICS].



## Computing Uncertainties: ASME Guidelines

### Recommended procedure (from J. Fluids Eng.)

1. Define a representative mesh size  $h$ , e.g.:

$$h = \left[ \frac{1}{N} \sum_{i=1}^N \Delta V_i \right]^{1/3}$$

2. Run simulations on 3 significantly different meshes, with refinement factor between each:

$$r = \frac{h_{coarse}}{h_{fine}} > 1.3$$

3. Calculate effective order of convergence  $p$
4. Calculate extrapolated values
5. Calculate fine "grid convergence index" (percent error):

$$GCI_{fine}^{21} = \frac{1.25}{r_{21}^p - 1} \left| \frac{u_1 - u_2}{u_1} \right|$$

**Figure 21.** Slide from Dr. Wagner <sup>9</sup> on GCI % error calculation

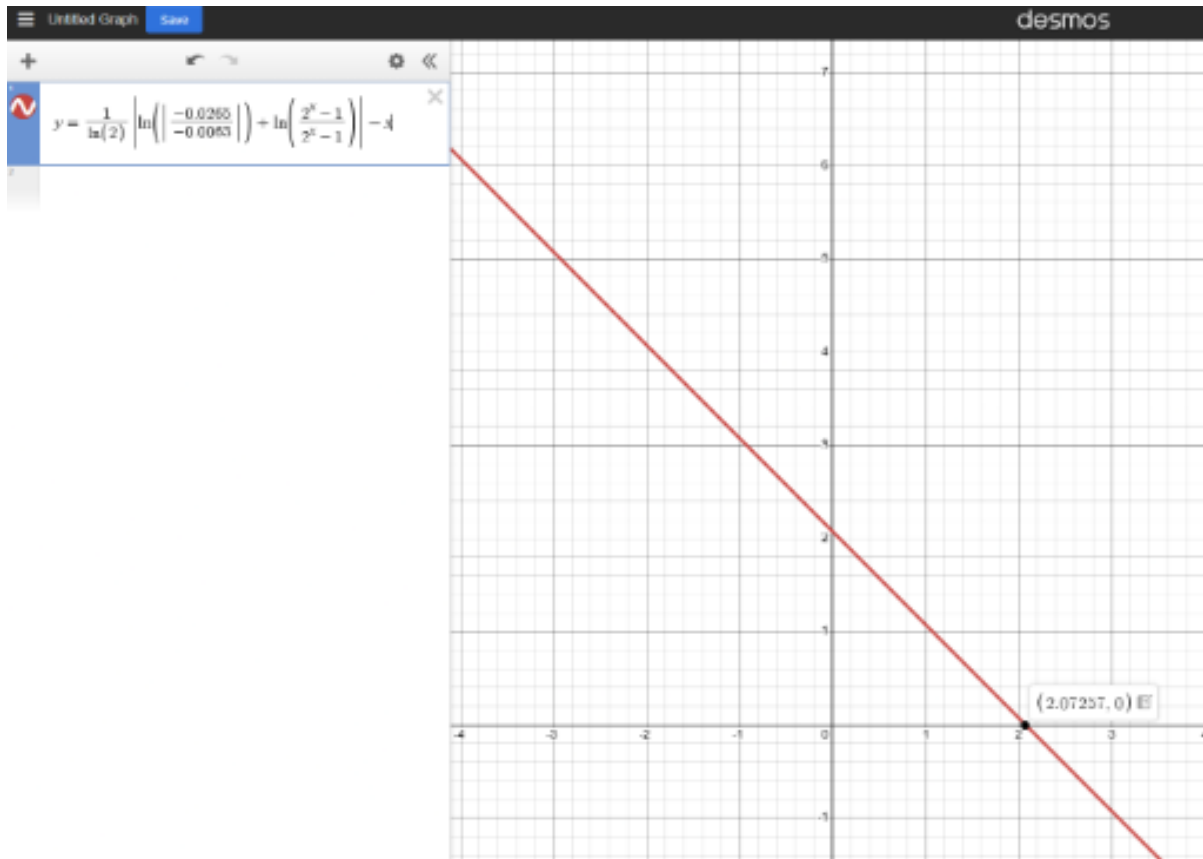
**Table 6.** Richardson extrapolation calculations. See link to spreadsheet <sup>10</sup> for explicit formulas.

|               |         |                 |         |
|---------------|---------|-----------------|---------|
| <b>u1</b>     | 0.1169  | <b>r_21</b>     | 2       |
| <b>u2</b>     | 0.1106  | <b>r_32</b>     | 2       |
| <b>u3</b>     | 0.0904  | <b>p</b>        | 2.07257 |
| <b>eps_21</b> | -0.0063 | <b>u_extrap</b> | 0.1189  |
| <b>eps_32</b> | -0.0265 | <b>GCI %</b>    | 2.09%   |
| <b>s</b>      | 1       | <b>% Error</b>  | 13.11%  |

<sup>9</sup> Wagner, G. J. (n.d.). *Lecture 5: Errors and Verification* [ME 378: APPLIED COMPUTATIONAL FLUID DYNAMICS].

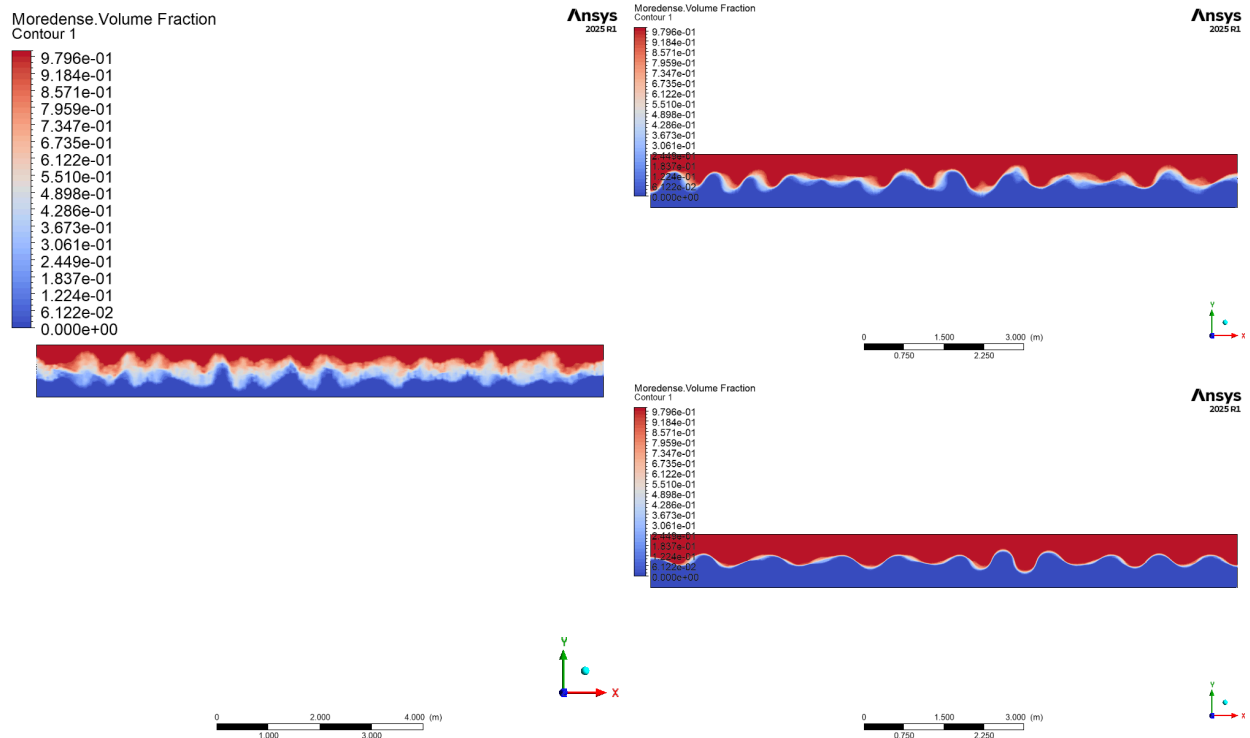
<sup>10</sup>

<https://docs.google.com/spreadsheets/d/1dCyAFShAjj2M1nuMrjusWI5sQ83mowf-ijZZOLN9N24/edit?usp=sharing>

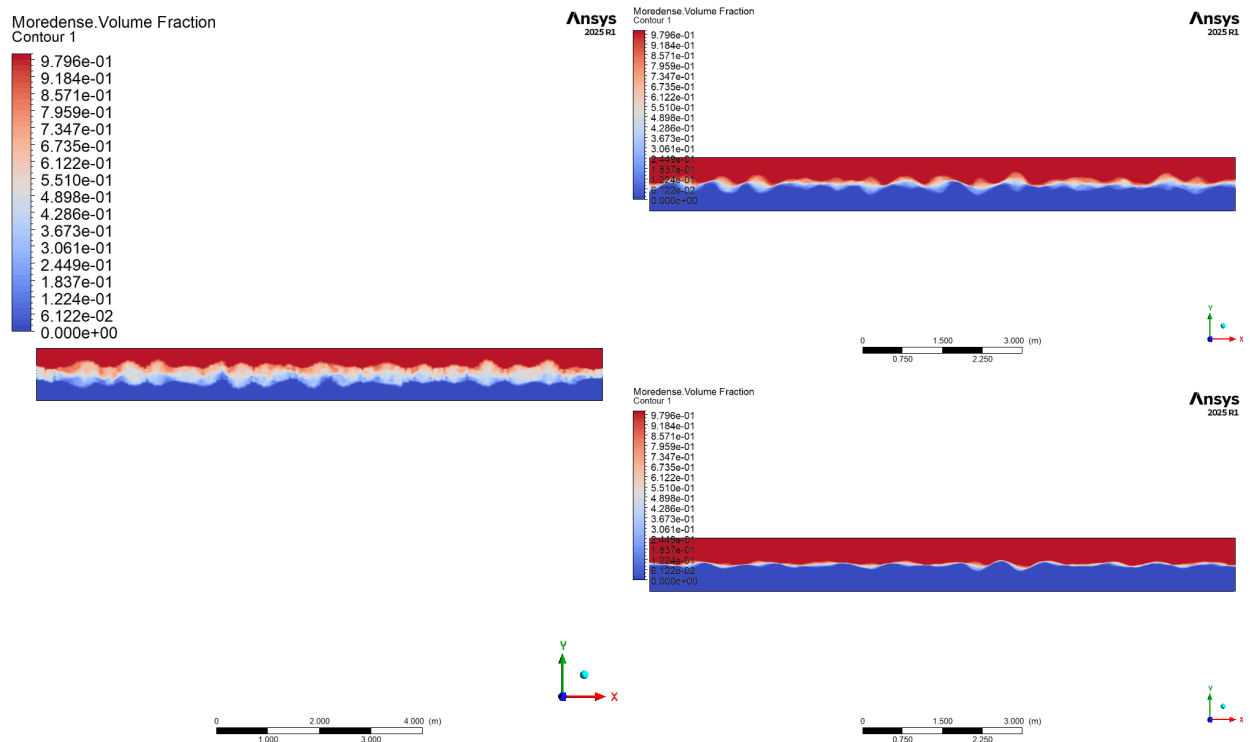


**Figure 22.** Using Desmos to solve for p-value (x in Desmos) in Richardson Extrapolation process.  
X-intercept

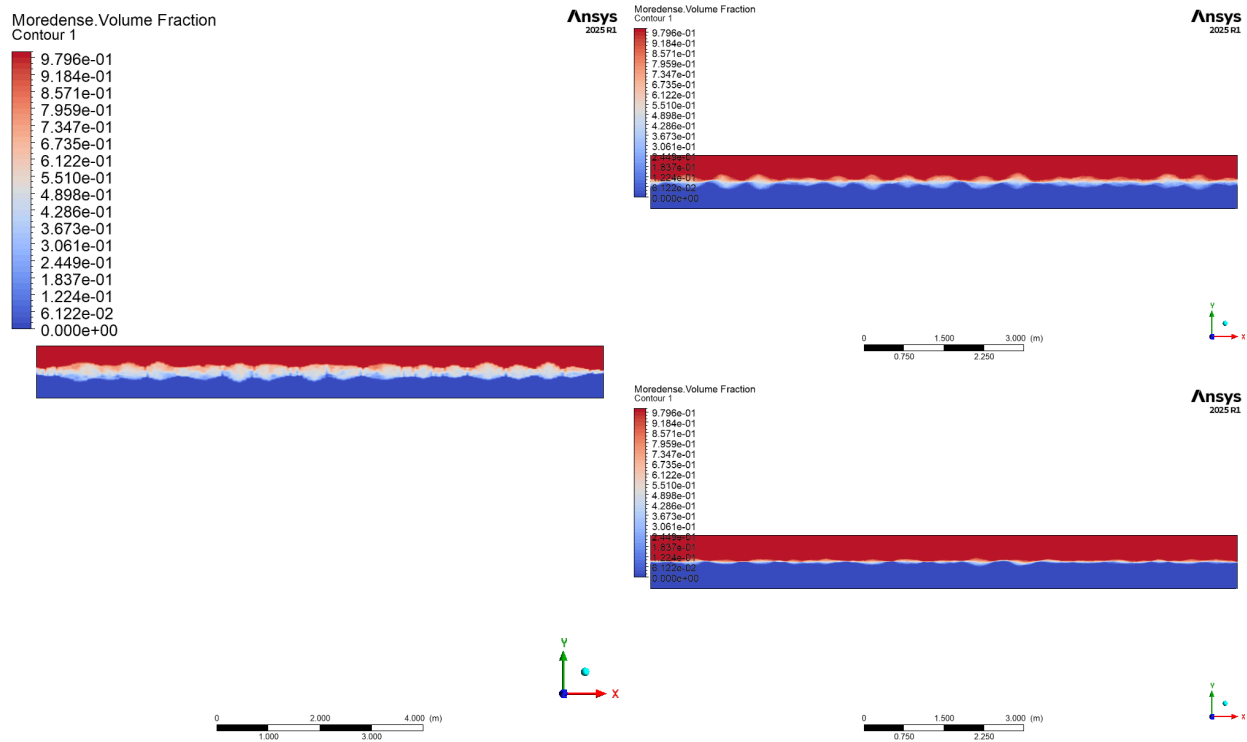
## Appendix D - 1000/999 Volume Fraction Contours at Various Times



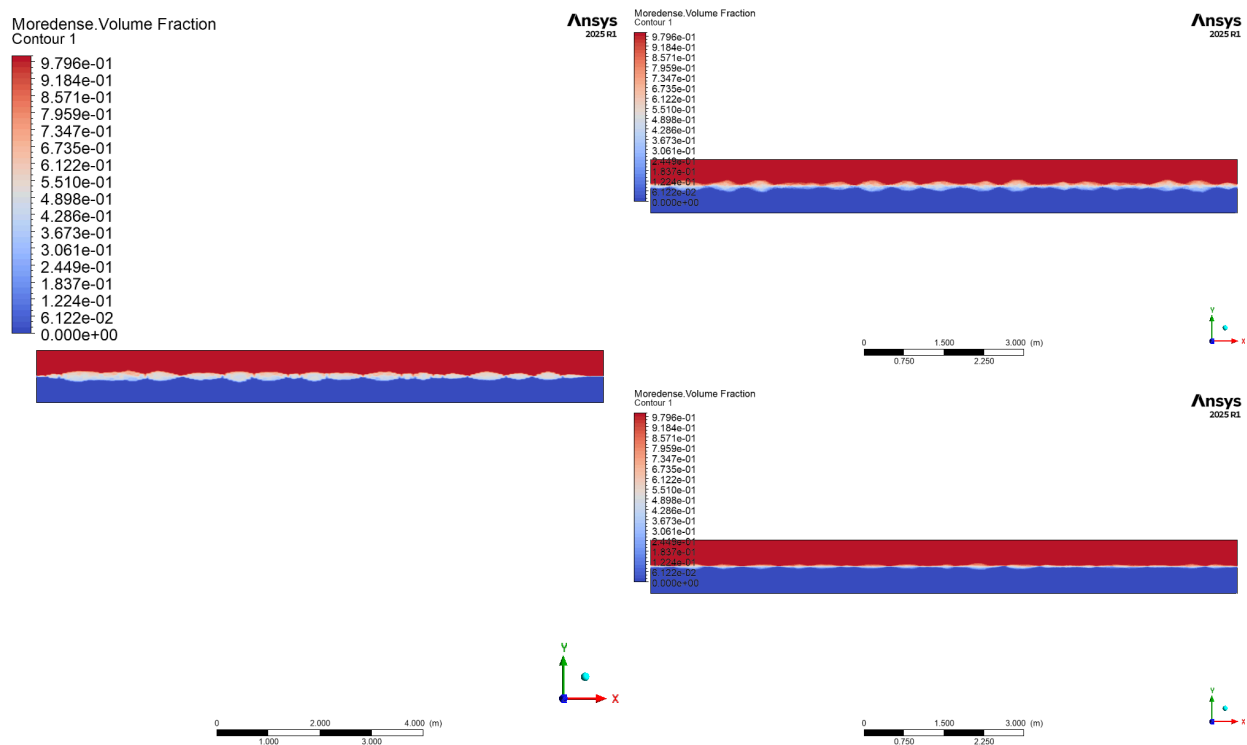
**Figure 23.** 1000/999 volume fraction contours at  $t=90$ s for coarse, medium, and fine meshes (from right to left, top to bottom.)



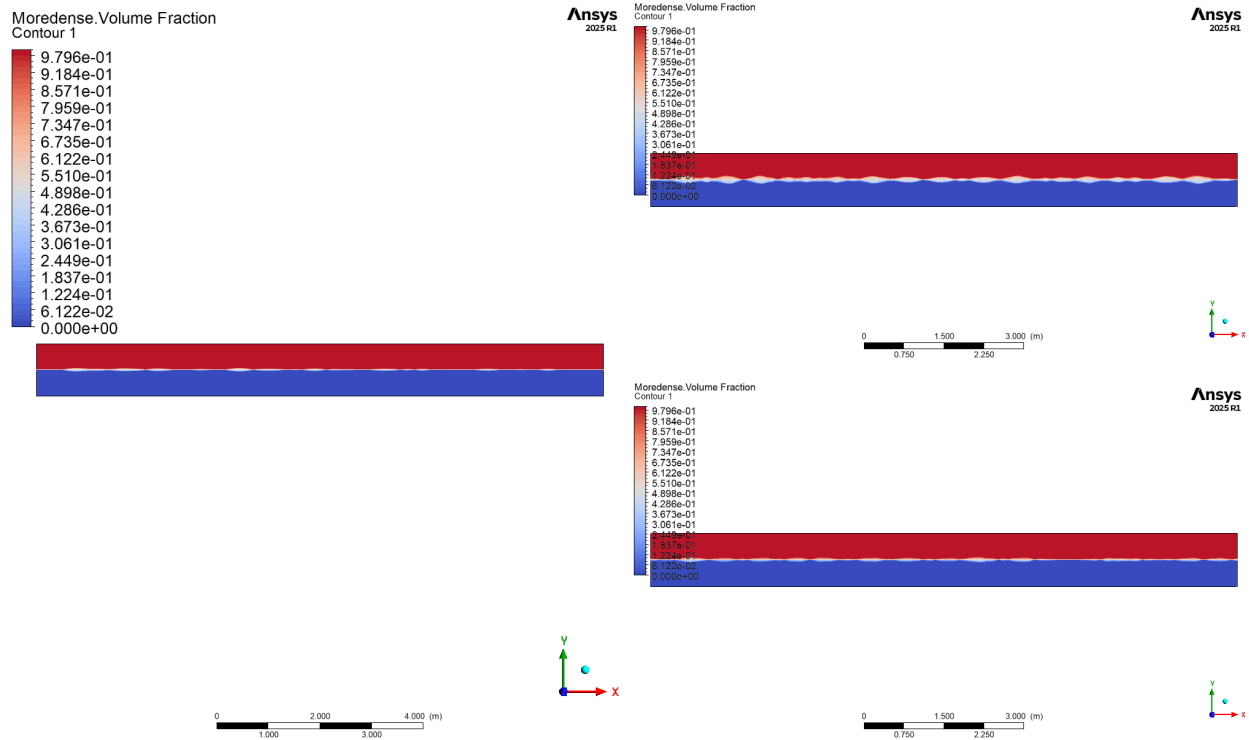
**Figure 24.** 1000/999 volume fraction contours at  $t=80s$  for coarse, medium, and fine meshes (from right to left, top to bottom.)



**Figure 25.** 1000/999 volume fraction contours at  $t=70s$  for coarse, medium, and fine meshes (from right to left, top to bottom.)



**Figure 26.** 1000/999 volume fraction contours at  $t=60s$  for coarse, medium, and fine meshes (from right to left, top to bottom.)



**Figure 27.** 1000/999 volume fraction contours at  $t=50s$  for coarse, medium, and fine meshes (from right to left, top to bottom.)

## Appendix E - Code for FFT in Python

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import savgol_filter
from scipy.optimize import curve_fit
from scipy.fft import rfft, rfftfreq # Import FFT functions

# Load the data
data = pd.read_csv("file_name.csv", delimiter=',', skiprows=5)
data = data.set_axis(['X', 'Y'], axis=1)
data = data.sort_values(by='X', ascending=True)

X = np.array(data["X"], dtype=float)
Y = np.array(data["Y"], dtype=float)

# Normalize X to start at 0 (helps with frequency fitting)
X -= X.min()

# Apply a Savitzky-Golay filter for smoothing
window_length = 101 # Must be odd
polyorder = 7
Y_smooth = savgol_filter(Y, window_length=window_length, polyorder=polyorder)

# **Estimate Initial Frequency Using FFT**
N = len(X)
T = (X.max() - X.min()) / (N - 1) # Approximate sampling interval

# Perform FFT
Y_fft = rfft(Y_smooth - np.mean(Y_smooth)) # Remove DC component before FFT
frequencies = rfftfreq(N, d=T)

cutoff_freq = 0 # Define cutoff frequency based on your signal characteristics
Y_fft[np.abs(frequencies) < cutoff_freq] = 0

# Find the dominant frequency (ignoring DC component at index 0)
dominant_freq_index = np.argmax(np.abs(Y_fft[1:])) + 1 # Avoid DC component
estimated_freq = frequencies[dominant_freq_index]

# Define the sine wave function
def sine_wave(x, A, f, phi, offset):
    return A * np.sin(2 * np.pi * f * x + phi) + offset

# Initial parameter guess
initial_guess = [
    (np.max(Y_smooth) - np.min(Y_smooth)) / 2, # Amplitude

```

```

    estimated_freq, # Frequency from FFT
    0, # Phase
    np.mean(Y_smooth) # Offset
]

# Fit the sine curve
params, covariance = curve_fit(sine_wave, X, Y, p0=initial_guess)

# Extract parameters
amplitude, frequency, phase, offset = params

# Compute fitted curve
Y_fit = sine_wave(X, amplitude, frequency, phase, offset)

Plot results
plt.figure(figsize=(10, 5))
plt.plot(X, Y, label="Original Data", alpha=0.5)
plt.plot(X, Y_smooth, label="Smoothed Data (Savitzky-Golay)", linestyle="--")
plt.plot(X, Y_fit, label="Fitted Sine Curve", linestyle="-", linewidth=2)
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.title("Sine Curve Fit with Improved Frequency Estimation (FFT)")
plt.show()

# Compute wavelength
wavelength = 1 / frequency if frequency != 0 else np.inf
print(f"Fitted Sine Equation:  $Y = \{amplitude:.3f\} * \sin(2 * \pi * \{frequency:.3f\} * X + \{phase:.3f\}) + \{offset:.3f\}$ ")
print(f"Amplitude: {amplitude:.10f}")
print(f"Frequency: {frequency:.3f} Hz")
print(f"Wavelength: {wavelength:.3f} units")
print(f"Phase: {phase:.3f}")
print(f"Offset: {offset:.3f}")

```